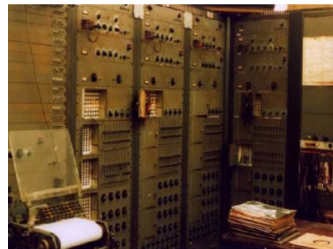


# Modeling the US Economy

- ▶ In 1949, economist Wassily Leontief worked with 250,000 data points produced by the United States BLS.
- ▶ Based on his research, he initially divided the US economy into 500 different sectors.
- ▶ One of the world's highest powered computers, Harvard's Mark II, could not process so much information.
- ▶ The model had to be further classified into a system of 42 equations of 42 unknowns.
- ▶ Such models are now known as **Leontief production models**.
- ▶ Mark II solved the system in 56 hours.



Based on *Linear Algebra and Its Applications* by D. Lay, et al.

Image Credit: <http://lasierrainformatica.blogspot.com/2013/06/el-harvard-mark-ii.html>

# Input-Output Model

- ▶ Instead of the 500 sectors Leontief identified, suppose the US economy is divided into 3 different sectors: Renewable Energy, Electricity, and Manufacturing.
- ▶ We can measure the total output for the year in each sector.
- ▶ We also know how each sector's output is divided among the other sectors.
- ▶ We can summarize this economy with an **input-output table**.

|                  |                            | Amount Purchased by |             |               |
|------------------|----------------------------|---------------------|-------------|---------------|
| Sector           | Total Output (in Billions) | Renewables          | Electricity | Manufacturing |
| Renewable Energy | 40                         | 10 (25%)            | 25 (62.5%)  | 5 (12.5%)     |
| Electricity      | 100                        | 7 (7%)              | 18 (18%)    | 75 (75%)      |
| Manufacturing    | 125                        | 20 (16%)            | 50 (40%)    | 55 (44%)      |

## Exchange Tables

We can convert the input-output table to an **exchange table** by giving the proportion of the output of each sector that is consumed by each of the other sectors.

| Proportion of Output from: |             |               |                  |
|----------------------------|-------------|---------------|------------------|
| Renewable Energy           | Electricity | Manufacturing | Purchased by     |
| 0.25                       | 0.07        | 0.16          | Renewable Energy |
| 0.625                      | 0.18        | 0.4           | Electricity      |
| 0.125                      | 0.75        | 0.44          | Manufacturing    |

- ▶ The total dollar amount of each sectors output is called the **price** of that sector.
- ▶ Thus,  $p_r$ ,  $p_e$ , and  $p_m$  denote total outputs of Renewable Energy, Electricity, and Manufacturing sectors, respectively.
- ▶ Leontief proved that there exists equilibrium prices that can be assigned to the total outputs for each sector in such a way that the income of each sector exactly balances its expenses
- ▶ How can we determine the equilibrium prices for each sector?

## Setting Up a System of Linear Equations

| Proportion of Output from: |             |               |                  |
|----------------------------|-------------|---------------|------------------|
| Renewable Energy           | Electricity | Manufacturing | Purchased by     |
| 0.25                       | 0.07        | 0.16          | Renewable Energy |
| 0.625                      | 0.18        | 0.4           | Electricity      |
| 0.125                      | 0.75        | 0.44          | Manufacturing    |

- ▶ Down each column is the proportion of that sector which is purchased by each of the other three sectors.
- ▶ Across each row we see for a given sector, what proportion of their inputs came from each sector's output.
- ▶ For the renewable energy sector, we therefore have the following linear equation:

output from renewables = (input from renewables) + (input from electricity) + (input from manufacturing)

$$p_r = 0.25p_r + 0.07p_e + 0.16p_m$$

$$0.75p_r - 0.07p_e - 0.16p_m = 0$$

# Setting Up a System of Linear Equations

| Proportion of Output from: |             |               |                  |
|----------------------------|-------------|---------------|------------------|
| Renewable Energy           | Electricity | Manufacturing | Purchased by     |
| 0.25                       | 0.07        | 0.16          | Renewable Energy |
| 0.625                      | 0.18        | 0.4           | Electricity      |
| 0.125                      | 0.75        | 0.44          | Manufacturing    |

Similarly deriving linear equations for the other sectors, we have the following system:

$$0.75p_r - 0.07p_e - 0.16p_m = 0$$

$$-0.625p_r + 0.82p_e - 0.4p_m = 0$$

$$-0.125p_r - 0.75p_e + 0.56p_m = 0$$

## Finding the Equilibrium for the Economy

We have the following augmented matrix associated to the system:

$$\left[ \begin{array}{ccc|c} 0.75 & -0.07 & -0.16 & 0 \\ -0.625 & 0.82 & -0.4 & 0 \\ -0.125 & -0.75 & 0.56 & 0 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|c} 1 & 0 & -0.28 & 0 \\ 0 & 1 & -0.70 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

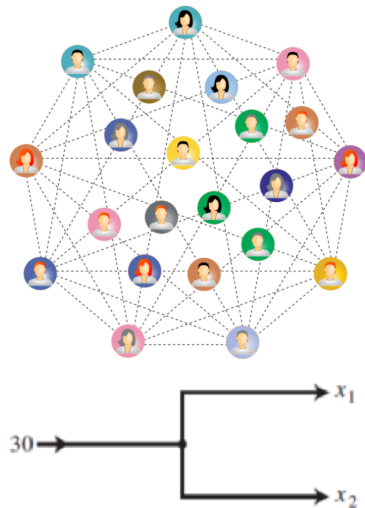
Notice  $p_m$  is a free variable, and we have equilibrium solution:

$$\mathbf{p} = \begin{bmatrix} p_r \\ p_e \\ p_m \end{bmatrix} = \begin{bmatrix} 0.28p_m \\ 0.7p_m \\ p_m \end{bmatrix} = p_m \begin{bmatrix} 0.28 \\ 0.7 \\ 1 \end{bmatrix}$$

- ▶ If this economy has  $p_m = 125$  billion dollars,
- ▶ Then if we want to ensure the economy is functioning at its equilibrium level (everything produced is used by other sectors):
  - ▶ Set  $p_r = (0.28)(125) = 35$  billion dollars, and
  - ▶ Set  $p_e = (0.7)(125) = 87.5$  billion dollars.

# Network Flow

- ▶ A **network** consists of a set of points, called **nodes** with lines, called **branches** connecting some or all of the nodes.
- ▶ The direction of the flow is indicated by each branch (are things flowing in or out of the node?).
- ▶ The flow amount (or rate) is either given or denoted by a variable.
- ▶ We assume the total flow into a network equals the total flow out of the network.
- ▶ The goal is to determine the flow in each branch when partial information is known.
- ▶ Network flows have applications to current flow through a circuit, flow of goods through supply chains, social networks, and **urban planning** to name a few.



# Traffic Flow in Baltimore

The network in the figure shows the flow of traffic (in vehicles per hour) over several one way streets in downtown Baltimore during a typical early afternoon. Determine the general flow pattern for the network.

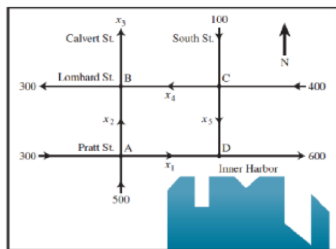


FIGURE 2 Baltimore streets.

| Intersection | Flow in     | Flow out      |
|--------------|-------------|---------------|
| A            | $300 + 500$ | $= x_1 + x_2$ |
| B            | $x_2 + x_4$ | $= 300 + x_3$ |
| C            | $100 + 400$ | $= x_4 + x_5$ |
| D            | $x_1 + x_5$ | $= 600$       |

$$\begin{array}{rcl}
 x_1 + x_2 & = & 800 \\
 x_2 - x_3 + x_4 & = & 300 \\
 x_4 + x_5 & = & 500 \\
 x_1 + x_5 & = & 600 \\
 x_3 & = & 400
 \end{array}$$

# Solving the System

We need to solve the following nonhomogeneous linear system of equations:

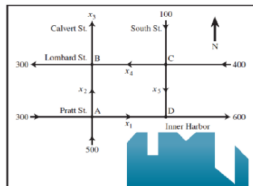


FIGURE 2 Baltimore streets.

$$\begin{array}{rclcrcl} x_1 & + & x_2 & & & = & 800 \\ & & x_2 & - & x_3 & + & x_4 & = & 300 \\ & & & & x_4 & + & x_5 & = & 500 \\ x_1 & & & & & + & x_5 & = & 600 \\ & & x_3 & & & & & = & 400 \end{array}$$

We have an augmented matrix

$$\left[ \begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 800 \\ 0 & 1 & -1 & 1 & 0 & 300 \\ 0 & 0 & 0 & 1 & 1 & 500 \\ 1 & 0 & 0 & 0 & 1 & 600 \\ 0 & 0 & 1 & 0 & 0 & 400 \end{array} \right] \rightarrow \left[ \begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 1 & 600 \\ 0 & 1 & 0 & 0 & -1 & 200 \\ 0 & 0 & 1 & 0 & 0 & 400 \\ 0 & 0 & 0 & 1 & 1 & 500 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \begin{cases} x_1 = 600 - x_5 \\ x_2 = 200 + x_5 \\ x_3 = 400 \\ x_4 = 500 - x_3 \\ x_5 \text{ is free} \end{cases}$$